CRT Chapter 6

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Question 2

Explain the difference between a method declaration and a method body.

- **Method Declaration (Header):** This is the top line of a method. It tells Java the method's name, who can access it, what type of data it returns, and what inputs (parameters) it needs.
- **Method Body:** This is the code inside the curly braces {} right after the header. It contains the actual steps and instructions that run when the method is called.

Question 9

Find and explain the error in the code.

Java

```
public class MethodCallExample {
   public static void main(String[] args) {
      int num;
      doSomething();
      num = doSomething();
   }

   public static int doSomething() {
      return(5);
   }
}
```

- **Explanation of Error:** The method doSomething() is written to send back a number (an int). On line 4, the code calls doSomething(); by itself without saving or using the number 5 that comes back. To fix this, you should either assign the return value to a variable or remove that extra standalone line.

Question 11: True or False

- **a) Breaking a task down into methods is called procedural abstraction.**
  - **Answer:** True
- **b) A method call consists of the method declaration in an assignment statement.**
  - **Answer:** False. A method call just uses the method's name and parentheses (like doSomething()), not the full declaration line.
- **c) A void method must return a value.**
  - **Answer:** False. A void method does not return any value.
- **d) An access modifier declares the return type of a method.**
  - **Answer:** False. An access modifier (like public or private) sets who can use the method, not what type of data it gives back.
- **e) The keyword static declares a method is a class method.**
  - **Answer:** True
- **f) Method parameters are enclosed by braces ({}).**
  - **Answer:** False. Parameters go inside parentheses (). Braces {} hold the body of the method.
- **g) Local variables can be used by any method in a class.**
  - **Answer:** False. A local variable can only be used inside the specific method where it was created.
- **h) The value of an argument passed to a method can be changed in an assignment statement in the method.**
  - **Answer:** False. Java passes values by copy. Changing the variable inside the method does not change the original value outside of it.
- **i) Method overloading means that an application contains more than 10 methods.**
  - **Answer:** False. Method overloading means having two or more methods with the exact same name but different parameters.
- **j) The return statement is used to send a value back to the calling statement.**
  - **Answer:** True
- **k) The precondition of a method states the data types of the method's parameters.**
  - **Answer:** False. A precondition is a rule that must be true *before* the method runs (for example, a number must be greater than zero).
- **l) The postcondition of a method describes the way the method accomplishes its task.**
  - **Answer:** False. A postcondition describes *what* the result is after the method finishes, not *how* it did the work.